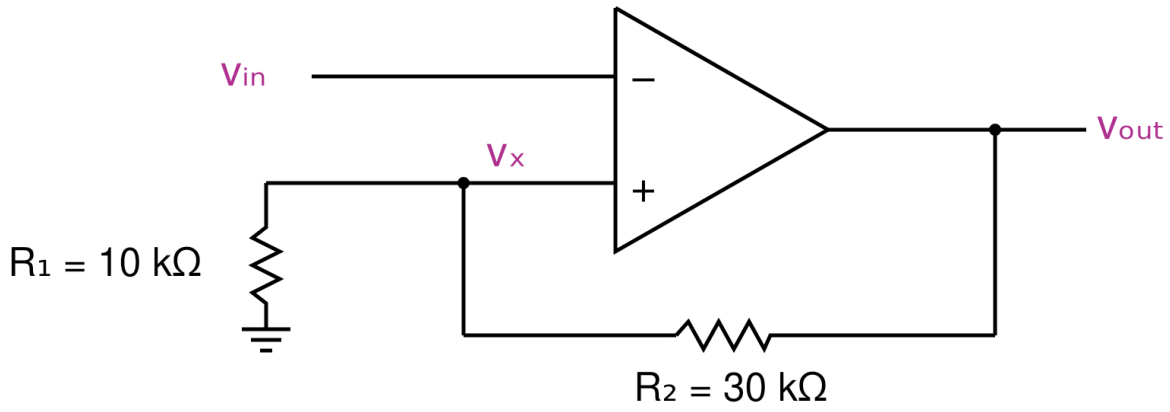
Tutorial question 4 from Week 5

The Schmitt trigger shown has output saturation levels of ± 12 V. Assume an ideal op amp.

(a) Find V_x in terms of v_{out}, and calculate the upper and lower switching thresholds.

(b) Initially, $v_{\text{in}} = 0$ V and $v_{\text{out}} = +12$ V. The input rises to +5 V and then returns to 0 V. Determine the output at each stage.

(c) If the input subsequently fluctuates between -1 V and +1 V , will the output switch? Explain briefly.

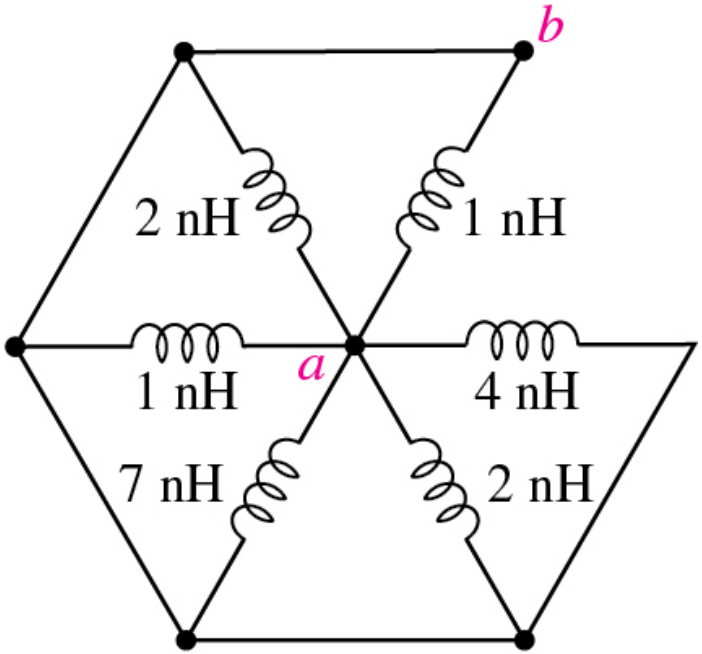


Tutorial question 1

A 100 mH inductor has voltage $v_L = 2e^{-3t}\text{V}$ across its terminals. Determine the resulting inductor current if $i_L(-0.5) = 1\text{A}$.

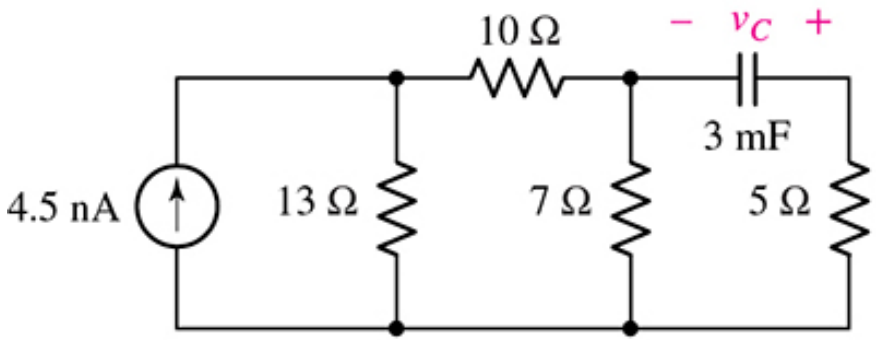
Tutorial question 2

Determine the equivalent inductance seen looking into the terminals marked a and b of the network represented below.

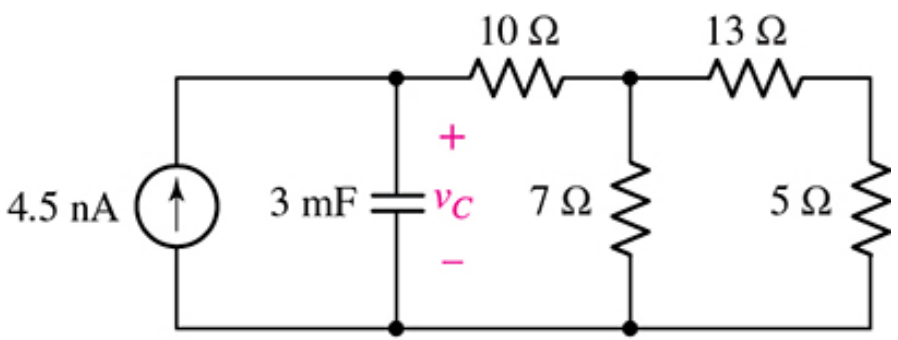


Tutorial question 3

For each circuit shown in Figure below, calculate the voltage labeled v_c.



(a)



(b)

Tutorial question 4 from Week 5

(a)

Since no current flows into the op-amp inputs, the voltage divider gives

$$V_x = \frac{R_1}{R_1 + R_2} v_{\text{out}} = \frac{10}{10 + 30} v_{\text{out}} = 0.25 v_{\text{out}} .$$

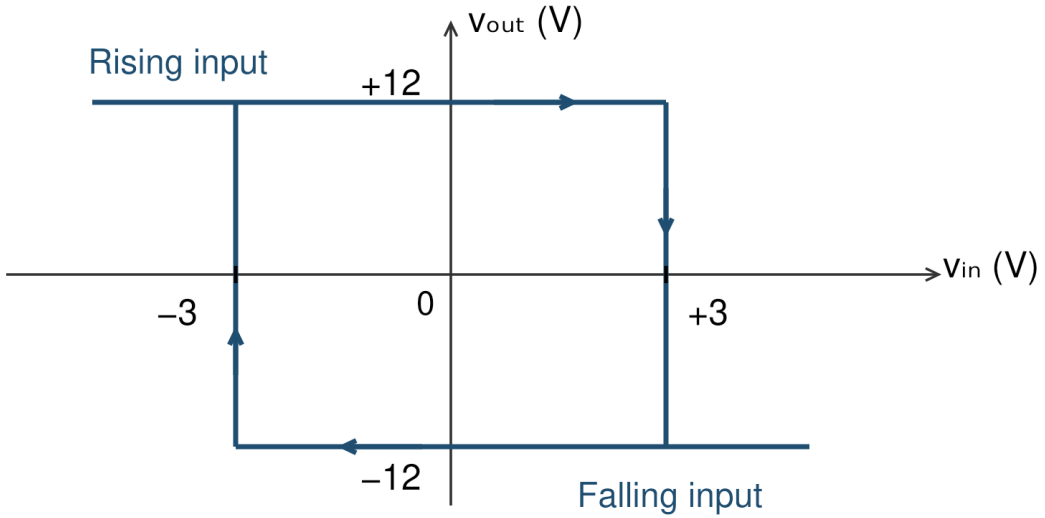
Therefore, the upper and lower switching thresholds are

$$V_T^{\text{upper}} = 0.25(12) = +3 \text{ V},$$
$$V_T^{\text{lower}} = 0.25(-12) = -3 \text{ V}.$$



Tutorial question 4 from Week 5

(b) As the input rises above +3 V , the output switches from +12 V to -12 V . The reference voltage V_x then becomes -3 V . When the input returns to 0 V , it remains above -3 V , so the output stays at -12 V .



(c) The output remains at -12 V . Input fluctuations between -1 V and +1 V do not cross the lower threshold of -3 V . This illustrates how hysteresis prevents unwanted switching caused by small input fluctuations.

Solution to Tutorial question 1

$$i(t) = \frac{1}{L}\int_{-0.5}^{t} v(t')dt' + i(t_0 = -0.5)$$

$$= 10\int_{-0.5}^{t} 2e^{-3t'}dt' + 1$$

$$= 10\left[-\frac{2}{3}e^{-3t} + \frac{2}{3}e^{1.5}\right] + 1$$

$$= -\frac{20}{3}e^{-3t} + 30.9$$



Solution to Tutorial question 2

All inductances are in parallel, so

$$\frac{1}{L_{eq}} = \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{1} + \frac{1}{7} + \frac{1}{2} + \frac{1}{4}\right) \times 10^9$$

Thus,

$$L_{eq} = 295\ \text{pH}$$

Solution to Tutorial Question 3

(a)

No current through 5 Ω resistor,

$$-v_C = (4.5 \times 10^{-9})(13)\frac{7}{13+10+7} = 13.65 \times 10^{-9}\text{V}$$

Thus,

$$v_C = -13.65 \times 10^{-9}\text{V}$$

(b)

v_C is equal to voltage across current source, so

$$v_C = (4.5 \times 10^{-9})(10 + 7||18) = 67.7 \times 10^{-9}\text{V}$$

